

INDR 100

INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #12 Productivity

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1

Agenda of the Day

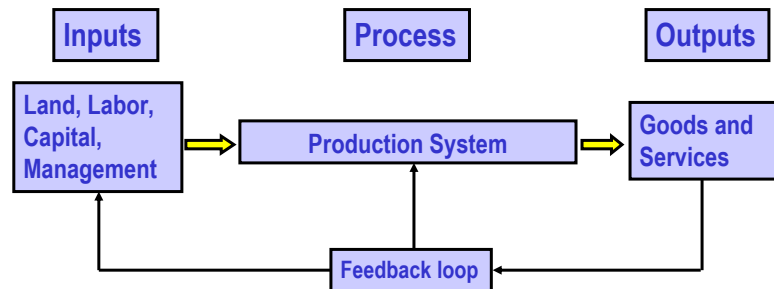
- Productivity
- Factors

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2

2

System: Transforming input to output

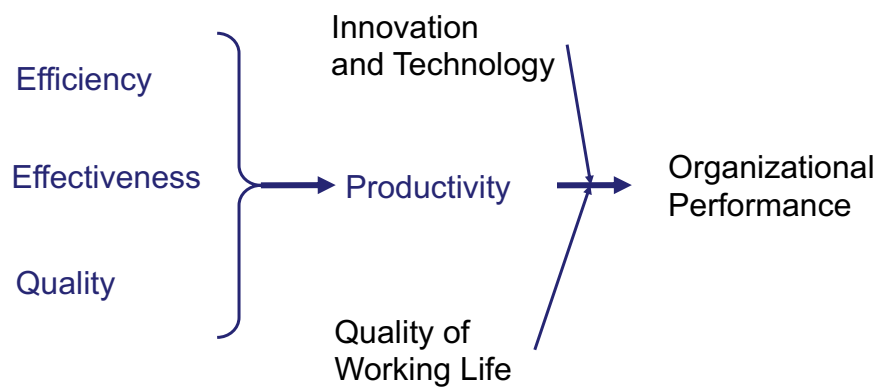


How to measure the performance of this transformation?

3

3

Total organization performance



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4

4

Productivity

- ❑ One of the most important responsibilities of an operations manager is to achieve **productive use of organization's resources**.
- ❑ **Productivity** is an index that measures how well the input (capital, labor, materials, energy, and other resources) are being used to produce the output (goods and services).
- ❑ It is usually expressed as the ratio of output to input:

$$Productivity = \frac{Output}{Input}$$

5

5

How do we use productivity?

- ❑ We can track an organization's performance over time.
- ❑ We can judge the performance of an entire industry or a country.

6

6

Production improvement does not necessarily mean productivity improvement. (Productivity $\neq$ being productive)

- ❑ Suppose a bank processed 1,000 checks yesterday, using 20 hours of labor.
- ❑ Let's say that the same bank processed 1,200 checks today, using 24 hours of labor.
- ❑ Production has increased by 20%, from 1,000 to 1,200 checks. However, the labor productivity for this operation is unchanged, because $1,000/20$ is equal to 50 checks per labor hour yesterday, and $1,200/24$ is equal to 50 checks per labor hour today.
- ❑ Therefore, improvement in production does not necessarily generate improvement in labor productivity.

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7

7

Recap

- ❑ Increase output by using the same or a lesser amount of (input) resource.
- ❑ Reduce amount of (input) resource used while keeping output constant or increasing it.
- ❑ Use more resource as long as output increases at a greater rate.
- ❑ Decrease output as long as resource use decreases at a greater rate.
- ❑ Production is concerned with the activity of producing goods and services.
- ❑ Productivity is concerned with the efficiency and effectiveness with which these goods and services are produced.

12

12

Efficiency and Effectiveness for productivity improvement

- ❑ Efficiency is a necessary but not a satisfactory condition for productivity. In fact, both effectiveness and efficiency are necessary in order to be productive.
- ❑ Efficiency is the ratio of actual output generated to the expected (or standard) output prescribed.
- ❑ Effectiveness, on the other hand, is the degree to which the relevant goals or objectives are achieved.
- ❑ Effectiveness involves first determining the relevant (right) goals or objectives and then achieving them.
 - If, for example, nine out of ten relevant goals are achieved, the effectiveness is 90%. One can be very efficient and still not be productive.

13

13

Efficiency improvement does not guarantee productivity improvement

- ❑ For example, suppose a dentist removes a patient's tooth in half the usual time and boasts (show off) his assistants:
 - “I have been twice as efficient as in the past.”
- ❑ The assistants, who view the situation say something differently:
 - “What a disaster – the dentist removed the wrong tooth!”
- ❑ Here, the dentist's effectiveness was zero because he did not achieve the relevant goal, even though his efficiency improved by 200%.

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14

14

Single Factor Approach to Measuring Productivity

- Capital - Number of products produced divided by asset value
- Materials - Number of products produced divided by dollars spent on materials consumed
- Direct Labor - Number of products produced divided by direct labor-hours
- Overhead - Number of products produced divided by dollars spent on operating cost
- **Single Factor measures:**

$$\frac{\text{Output}}{\text{Labor}}$$

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$$\frac{\text{Output}}{\text{Materials}}$$

$$\frac{\text{Output}}{\text{Capital}}$$

$$\frac{\text{Output}}{\text{Overhead}}$$

15

15

Multi Factor Measures of Productivity

Combination of some of the single factors discussed:

- **Multifactor measures**

$$\frac{\text{Output}}{\text{Labor} + \text{Machine}}$$

$$\frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Energy}}$$

- **Total measure**

$$\frac{\text{Goods or Services Produced}}{\text{All inputs used to produce them}}$$

More measures to be defined in
INDR 471 Service Operations Analysis

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16

16

Typical Measures of Productivity in Different Organizations

Restaurant	Customers (meals) per labor hour
Retail store	Sales per square meter
Chicken farm	Kgs of meat per kg of feed
Utility plant	Kilowatts per ton of coal
Paper mill	Tons of paper per cubic meter of wood

17

17

Examples of Production Systems

OPERATION	BANK	RESTAURANT	HOSPITAL	UNIVERSITY	MANUFACTURING PLANT	AIRLINE
INPUT	Tellers, staff, computer equipment, facilities, and energy 	Cooks, waitresses, food, equipment, facilities, and energy 	Doctors, nurses, staff, equipment, facilities, and energy 	Faculty, staff, equipment, facilities, energy, and knowledge 	Equipment, facilities, labor, energy, raw materials 	Planes, facilities, pilots, flight attendants, maintenance people, labor, energy 
OUTPUT	Financial Services (loan deposits, safekeeping, accounts handled).	Meals, entertainment, satisfied customers	Health services (recovered patients, number of operations performed)	Educated students, research output, public service	Finished goods	Number of flights, number of customers carried
Measures of Productivity		Customers (meals) per labor hour			Paper Mill: Tons of paper per cubic meter of wood	

18

18

What Affects Labor Productivity?

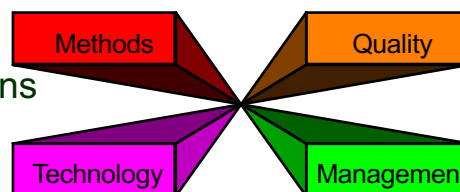
- ❑ Physical work environment
 - Technology, equipment, materials, lighting, layout
- ❑ Product quality
 - Defects, scrap, rework
- ❑ Employee job performance
 - Employee ability, motivation

19

19

Factors Affecting Productivity

- Standardization
- Technology
- Use of Internet, fax machines, e-mail, computerized billing, software
- Shortage of IT workers and other technical workers
- Equipment breakdowns
- Scrap rates
- Safety



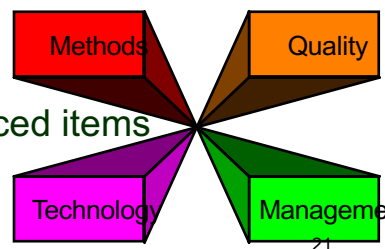
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20

20

Factors Affecting Productivity (cont'd)

- Methods
- Incentive plans that reward productivity
- Inadequate investment in training and education of the employees
- Labor turnover, layoffs, new workers
- Layout
- Design of the workspace
- Searching for lost or misplaced items
- Bottlenecks



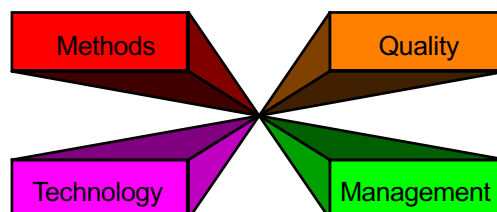
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21

21

Factors Affecting Productivity (cont'd)

- Inventory
- Capacity utilization
- Scheduling
- Part and material shortages
- Location



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22

22

Key Steps for Improving Productivity

- Develop productivity measures for all operations
- Determine critical (bottleneck) operations
- Develop methods for productivity improvements
- Establish reasonable goals
- Get management support (make it clear that management supports and encourages productivity improvements.)
- Measure and publicize improvements
- Invest on labor force by training and education

(Don't confuse productivity with efficiency)

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23

23

Continuous Improvement

- ❑ The aim is to **make incremental changes** in the system over time.
 - understand fundamental issues in the system.
 - observe and evaluate the system.

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24

24

How to observe/evaluate a system?

- ❑ Work environment is a major component of the system.
- ❑ **Work-design, work measurement, process activities, and reward systems** are the elements of the work environment, which help IEs to understand such fundamental issues.
- ❑ IEs can recognize the bottlenecks in the system, identify unnecessary and repetitive tasks, and eliminate them in the improved system.

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25

25

Tools and methods helping to observe/evaluate a system

- ❑ **Process analysis:** Charts and diagrams
- ❑ **Assembly line:** Workers have to perform very simple tasks repetitively → an essential method for an efficient production line
- ❑ How about **worker satisfaction?**

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26

26

Tools and methods helping to observe/evaluate a system

- Work design:
 - Job design: Method study (job simplification)
 - Work measurement:
 - Time study (time standards)
 - Work sampling
 - Worker compensation

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27

27

Improving Productivity in a Multi-Input Production Environment

A factory produces **metal components** used in electric scooters. The production system operates 2 shifts per day, 5 days a week.

Weekly Inputs

- Labor: 12 workers × 40 hrs = **480 labor-hours**
- Material cost = **40,000 TL**
- Energy cost = **12,000 TL**
- Machine hours = **600 machine-hours**
- Labor cost = 180 TL/hr
- Machine cost = 50 TL/hr

Output = 3,000 components

Total Input Cost:

$$(480 \times 180) + 40,000 + 12,000 + (50 \times 600) = \boxed{168,400 \text{ TL}}$$

29

Base-case productivity

Labor Productivity

$$\square \frac{3000}{480} = \boxed{6.25 \text{ components per labor-hour}}$$

Machine Productivity

$$\square \frac{3000}{600} = \boxed{5 \text{ components per machine-hour}}$$

Multifactor Productivity (MFP)

$$\square \frac{3000}{168,400} = \boxed{0.01782 \text{ components per TL}}$$

30

Class Exercise

Management is evaluating three independent improvement options:

- **Scenario A — Reduce Machine Downtime**

 - A maintenance overhaul costs **20,000 TL** (one-time, ignore for weekly productivity).
 - Downtime reduces from **15% to 5%**.
- **Scenario B — Using higher-quality materials:**

 - Material cost increases by **20%**
 - Rework and scrap decrease, increasing output by **18%**
 - Machine and labor hours stay the same
 - Energy cost decreases by **8%** due to reduced reprocessing
- **Scenario C — Worker Training + Automation Assist**

 - training program improving worker efficiency (output ↑ **10%**)
 - Automation assists reducing machine hours by **12%**
 - Labor hours remain the same
 - Material cost unchanged
 - Energy cost reduced by **5%**
 - Machine-hour cost still applies to the reduced machine-hour

31

Class Exercise

Management is evaluating three independent improvement options:

- ❑ **Scenario A — Reduce Machine Downtime**
 - A maintenance overhaul costs **20,000 TL** (one-time, ignore for weekly productivity).
 - Downtime reduces from **15% to 5%**.
- Compute new output.
- Recompute machine productivity and MFP.
- Compare the percentage improvement in MFP vs. Base Case:

Weekly Inputs

- ❑ Labor: 12 workers × 40 hrs = **480 labor-hours**
- ❑ Material cost = **40,000 TL**
- ❑ Energy cost = **12,000 TL**
- ❑ Machine hours = **600 machine-hours**
- ❑ Labor cost = 180 TL/hr → **86,400 TL**
- ❑ Machine cost = 50 TL/hr → **30,000 TL**
- Output = 3,000 components**

32

Solution-Scenario A

Downtime improves: 15% → 5%

- ❑ Productive machine-hours:

$$\frac{600}{0.85} \times 0.95 = 670.59$$

Machine Productivity

- ❑ If machining hours is the bottleneck the machine productivity does not change
- ❑ If machining hours is not bottleneck

$$\frac{3000}{670.59} = \boxed{4.47}$$

MFP

- ❑ If machining hours is the bottleneck, the total cost increases for machining hours, material and energy.

$$\text{Production} = 3000 \times \frac{0.95}{0.85} = 3353.94$$

$$\text{Cost} = (480 \times 180) + 40,000 \frac{0.95}{0.85} + 12,000 \times \frac{0.95}{0.85} + \left(50 \times 600 \times \frac{600}{0.85} \right) = 178047.1$$

$$\text{MFP} = \frac{3353.94}{178047.1} = 0.01884$$

$$\text{MFP Improvement} = 0.01884 - 0.0173 = 0.00154$$

- ❑ If machining hours is not bottleneck MFP does not change

33

Class Exercise

Management is evaluating three independent improvement options:

- ❑ **Scenario B — Using higher-quality materials:**
 - Material cost increases by **20%**
 - Rework and scrap decrease, increasing output by **18%**
 - Machine and labor hours stay the same
 - Energy cost decreases by **8%** due to reduced reprocessing
- Compute new output.
- Recompute MFP.
- Compare the percentage improvement in MFP vs. Base Case:

Weekly Inputs

- ❑ Labor: 12 workers × 40 hrs = **480 labor-hours**
- ❑ Material cost = **40,000 TL**
- ❑ Energy cost = **12,000 TL**
- ❑ Machine hours = **600 machine-hours**
- ❑ Labor cost = 180 TL/hr → **86,400 TL**
- ❑ Machine cost = 50 TL/hr → **30,000 TL**
- Output = 3,000 components**

34

Solution-Scenario B

What changes:

- ❑ Material cost ↑ 20% →

$$40,000 \times 1.20 = \boxed{48,000 \text{ TL}}$$
- ❑ Energy cost ↓ 8% →

$$12,000 \times 0.92 = \boxed{11,040 \text{ TL}}$$
- ❑ Output ↑ 18% →

$$3000 \times 1.18 = \boxed{3540 \text{ components}}$$
- ❑ Labor & machine hours unchanged.

Total Input Cost

- ❑ $86,400 + 48,000 + 11,040 + 30,000 = \boxed{175,440 \text{ TL}}$

MFP

- ❑ $\frac{3540}{175,440} = \boxed{0.02017}$

MFP Improvement

- ❑ $\frac{0.02017 - 0.01782}{0.01782} = \boxed{13.2\% \text{ increase}}$

35

Class Exercise

Management is evaluating three independent improvement options:

- ❑ **Scenario C — Worker Training + Automation Assist**
 - training program improving worker efficiency (output ↑ 10%)
 - Automation assists reducing machine hours by 12%
 - Labor hours remain the same
 - Material cost unchanged
 - Energy cost reduced by 5%
 - Machine-hour cost still applies to the reduced machine-hour
- Compute new labor productivity, machine productivity, and MFP.

Weekly Inputs

- ❑ Labor: 12 workers × 40 hrs = **480 labor-hours**
- ❑ Material cost = **40,000 TL**
- ❑ Energy cost = **12,000 TL**
- ❑ Machine hours = **600 machine-hours**
- ❑ Labor cost = 180 TL/hr → **86,400 TL**
- ❑ Machine cost = 50 TL/hr → **30,000 TL**
- Output = 3,000 components**

36

Solution-Scenario C

What changes:

- ❑ Output ↑ 10% →

$$3000 \times 1.1 = \boxed{3300 \text{ components}}$$

- ❑ Machine hours ↓ 12% →

$$600 \times 0.88 = \boxed{528 \text{ hours}}$$

- ❑ Energy Costs ↓ 5% →

$$12000 \times 0.95 = \boxed{11,400 \text{ TL}}$$

Total Input Cost

$$86,400 + 40,000 + 11,400 + 26,400 = \boxed{164,200 \text{ TL}}$$

Labor Productivity

$$\frac{3300}{480} = \boxed{6.875}$$

Machine Productivity

$$\frac{3300}{528} = \boxed{6.25}$$

MFP

$$\frac{3300}{164,200} = \boxed{0.02009}$$

MFP Improvement

$$\frac{0.02009 - 0.01782}{0.01782} = \boxed{12.7\%}$$

37

Remarks

□ Which scenario is best?

- If the company is cash-constrained,
- If the goal is maximum output growth,
- If energy efficiency is a priority,

Would you recommend combining scenarios? What trade-offs arise?

38

STUDY QUESTIONS

39

Example 1

7040 Units Produced

Sold for \$1.10/unit

Cost of labor : \$1,000

Cost of materials: \$520

Cost of overhead: \$2000

*Which productivity
measures can be
calculated?*

*What is the
multifactor
productivity?*

2-40

40

Solution 1

$$\text{MFP} = \frac{\text{Output}}{\text{Labor} + \text{Materials} + \text{Overhead}}$$

$$\text{MFP} = \frac{(7040 \text{ units}) * (\$1.10)}{\$1000 + \$520 + \$2000}$$

$$\text{MFP} = 2.20$$

2-41

41

Example 2

5,500 Units Produced

Sold for \$35/unit

500 labor hours are used

Cost of labor: \$25/hr

Cost of raw material: \$5,000

Cost of overhead: 2 x labor cost

What is the labor productivity?

2-42

What is the multifactor productivity?

42

Solution 2: Labor Productivity

➤ $5,500 \text{ units} / 500 \text{ hours} = 11 \text{ units/hour}$

Or we can arrive at a unitless figure:

➤ $(5,500 \text{ units} * \$35/\text{unit}) / (500 \text{ hours} * \$25/\text{hr}) = 15.4$

2-43

43

Solution 2: Multifactor Productivity

$$\text{MFP} = \frac{\text{Output}}{\text{Labor} + \text{Materials} + \text{Overhead}}$$

$$\text{MFP} = \frac{(5,500 \text{ units}) * (\$357/\text{unit})}{(500 \text{ hrs.}) * \$25/\text{hr.} + (\$5000) + 2 * (500 \text{ hrs.}) * \$25/\text{hr.}}$$

$$\text{MFP} = 4.52$$

2-44

44

Example 3

- You have just determined that your service employees have used a total of 2400 hours of labor this week to process 560 insurance forms. Last week the same crew used only 2000 hours of labor to process 480 forms.
- Which productivity measure should be used?
 - Answer: Could be classified as a Partial Measure.
- Is productivity increasing or decreasing?
 - Answer: Last week's productivity = $480/2000 = 0.24$, and this week's productivity is = $560/2400 = 0.23$. So, productivity has decreased slightly.

2-45

45